



# BRIGMAN

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1600 Pennsylvania Ave NW  
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Brigman Deep Sea Technology

Analysis Phase Debriefing

## BCIC Analysis Phase Debriefing Report

### Abyss Navigator Suite Project

#### *Lessons Learned and Process Improvement Recommendations*

February 7, 2026

## Distribution

**To:** Senior Management, Engineering Department Heads  
**From:** Requirements Analysis Team  
**Date:** February 7, 2026  
**Subject:** BCIC Methodology Lessons Learned and Process Improvements

## Executive Summary

This debriefing report provides a comprehensive review of the two-month BCIC analysis phase for the Abyss Navigator Suite project. The report captures lessons learned, process effectiveness evaluations, and recommendations for methodology improvement based on our vendor-led application of the BCIC framework.

Key insights from the debriefing include:

- BCIC methodology effectively identified 26.3% requirement incongruities early
- Visual modeling techniques reduced clarification time by approximately 40%
- Stakeholder indecision was the primary cause of schedule delays
- The IRC checklist proved highly effective for systematic requirement validation





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- Vendor-customer relationship was significantly strengthened through structured engagement

**Primary Recommendation:** Formalize the BCIC methodology as standard practice for all complex systems projects.

## 1 Project Context and Analysis Approach

### 1.1 Project Overview

The Abyss Navigator Suite represents Brigman's most complex autonomous systems project to date. As Triton Deepwater's selected vendor, we applied our BCIC methodology to transform ambiguous customer requirements into actionable specifications.

### 1.2 BCIC Methodology Implementation

Our team of five specialists implemented the four-phase BCIC approach:

- **Breakdown:** Systematic deconstruction using Incongruous Requirements Checklist
- **Clarify:** Stakeholder engagement with visual modeling support
- **Interpret:** Translation to IEEE 830 compliant requirements
- **Categorize:** Organization into SRS-ready structure

## 2 Breakdown Activity Debrief

### 2.1 Process Effectiveness

The IRC checklist proved highly effective, identifying 15 incongruities across 57 requirements. The four validation lenses (ambiguity, testability, consistency, completeness) pro-





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vided comprehensive coverage.

## 2.2 What Worked Well

- Early identification of critical safety and communication gaps
- Quantitative measurement of requirement quality
- Systematic approach ensured no requirement was overlooked
- Effective prioritization of clarification needs

## 2.3 Areas for Improvement

- **Timeline Estimation:** Underestimated time required for high-ambiguity documents
- **Tool Support:** Manual IRC application was labor-intensive
- **Stakeholder Preparation:** Customer needed more guidance on requirement specificity

## 2.4 Recommendations

1. Develop automated IRC tool support for future projects
2. Create stakeholder education materials on requirement writing
3. Refine timeline estimation formula based on ambiguity metrics





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### 3 Clarify Activity Debrief

#### 3.1 Process Effectiveness

Three structured stakeholder sessions resolved 86.7% of identified incongruities. Visual modeling was particularly effective in bridging communication gaps.

#### 3.2 What Worked Well

- Visual models facilitated breakthrough understanding
- Structured sessions yielded concrete decisions
- Comprehensive documentation ensured traceability
- Relationship building strengthened customer partnership

#### 3.3 Areas for Improvement

- **Decision Latency:** Stakeholder indecision on complex items
- **Organizational Silos:** Conflicting department priorities
- **Technology Anxiety:** Customer uncertainty about autonomous systems

#### 3.4 Recommendations

1. Implement decision deadlines for stakeholder responses
2. Conduct cross-departmental alignment workshops earlier
3. Develop technology education sessions for non-technical stakeholders





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## 4 Interpret Activity Debrief

### 4.1 Process Effectiveness

Transformation of ambiguous statements into 79 testable requirements, including 22 new atomic requirements addressing implicit needs.

### 4.2 What Worked Well

- Quantitative transformation of subjective statements
- Technical feasibility ensured by operational expertise
- IEEE 830 compliance established verification basis
- Bidirectional traceability maintained throughout

### 4.3 Areas for Improvement

- **Over-Specification:** Early interpretations included unnecessary details
- **Validation Bottlenecks:** Technical reviews created workflow delays
- **Terminology Consistency:** Multiple iterations required

### 4.4 Recommendations

1. Develop requirement interpretation templates
2. Implement parallel validation processes
3. Create standardized terminology dictionary





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## 5 Categorize Activity Debrief

### 5.1 Process Effectiveness

Organization of requirements into six logical categories enabling parallel SRS development and implementation planning.

### 5.2 What Worked Well

- Logical grouping supported parallel workstreams
- Standardized coding (SYS-, SAF-, COM-) enabled consistent tracking
- Categorization informed resource allocation and scheduling
- Quality assurance through requirement distribution analysis

### 5.3 Areas for Improvement

- **Framework Complexity:** Initial scheme was too granular
- **Dependency Management:** Cross-category dependencies required careful handling
- **Manual Processes:** Labor-intensive categorization

### 5.4 Recommendations

1. Develop categorization framework templates for different project types
2. Implement dependency tracking tools
3. Automate categorization where possible





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## 6 Visual Modeling Effectiveness Assessment

### 6.1 Impact Analysis

Visual modeling techniques proved critical to analysis success:

- **40% reduction** in clarification time with visual aids
- **Breakthrough understanding** in complex technical areas
- **Stakeholder alignment** achieved through shared visual reference points

### 6.2 Most Effective Models

1. State transition diagrams for safety workflows
2. Data flow diagrams for interface clarification
3. Context diagrams for system boundary definition
4. Use case diagrams for operational scenario validation

### 6.3 Recommendations

1. Develop standard visual model library for reuse
2. Train all analysts in visual modeling techniques
3. Integrate modeling tools with requirements management systems





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## 7 Risk Management Process Evaluation

### 7.1 Effectiveness Assessment

Early risk identification through BCIC analysis enabled proactive mitigation:

- All critical technical risks identified during Breakdown
- Mitigation strategies developed during Clarify
- Implementation risks addressed before SRS development

### 7.2 Process Improvements Needed

- Formal risk quantification methodology
- Integration with enterprise risk management systems
- Regular risk review cadence throughout project lifecycle

## 8 Business Value Assessment

### 8.1 Quantifiable Benefits

| Benefit Category                                  | Estimated Value |
|---------------------------------------------------|-----------------|
| Cost avoidance through early issue detection      | \$350,000+      |
| Schedule reduction through clear requirements     | 4-6 weeks       |
| Risk reduction through proactive mitigation       | 60-70%          |
| Quality improvement through systematic validation | 40-50%          |
| Customer relationship strengthening               | Strategic value |

Table 1: BCIC Analysis Business Value







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## 8.2 Strategic Value

- **Competitive Differentiation:** Systematic analysis capability
- **Methodology Validation:** Proven approach for complex projects
- **Knowledge Assets:** Reusable frameworks and templates
- **Team Capability:** Enhanced skills and experience

## 9 Process Improvement Recommendations

### 9.1 Immediate Actions (Next 30 Days)

1. Formalize BCIC methodology in corporate process documentation
2. Develop automated IRC tool prototype
3. Create visual modeling template library
4. Train additional teams in BCIC application

### 9.2 Medium-Term Improvements (3-6 Months)

1. Integrate BCIC with requirements management tools
2. Develop metrics dashboard for analysis effectiveness
3. Create customer education materials on requirement quality
4. Establish BCIC community of practice

### 9.3 Long-Term Enhancements (6-12 Months)

1. AI-assisted incongruity detection tools





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2. Real-time collaboration platforms for stakeholder engagement
3. Predictive analytics for requirement risk assessment
4. Industry certification for BCIC methodology

## 10 Team Performance Assessment

### 10.1 Individual Contributions

- **Project Lead:** Effective stakeholder coordination and methodology oversight
- **Safety Engineering:** Critical identification of emergency protocol gaps
- **Communications:** Quantitative definition of bandwidth constraints
- **Autonomous Systems:** Clarification of autonomy level definitions
- **Data Management:** Establishment of traceability frameworks

### 10.2 Team Dynamics

- **Strengths:** Cross-functional collaboration, systematic approach, customer focus
- **Areas for Development:** Timeline management, tool utilization, stakeholder education
- **Overall Rating:** Excellent - exceeded expectations for complex analysis

## 11 Conclusion and Next Steps

The BCIC analysis phase has demonstrated significant value through systematic requirement validation and stakeholder alignment. The methodology proved particularly effective





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for complex, innovative projects with high ambiguity levels.

### 11.1 Key Success Factors

1. Structured approach with clear phase transitions
2. Visual modeling for stakeholder engagement
3. Cross-functional team composition
4. Proactive risk identification and mitigation
5. Comprehensive documentation and traceability

### 11.2 Final Recommendations

1. Adopt BCIC as standard methodology for all complex projects
2. Invest in tool support for methodology implementation
3. Develop organizational capability through training and certification
4. Continuously refine methodology based on lessons learned

## DEBRIEFING COMPLETION CERTIFICATION





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### Analysis Phase Debriefing

| Name                        | Role                         | Signature |
|-----------------------------|------------------------------|-----------|
| Muhammad Ridwan Putra Jasni | Project Lead                 | _____     |
| Chng Zong Han               | Safety Engineer-<br>ing Lead | _____     |
| Lin Yuhao                   | Communications<br>Lead       | _____     |
| Muhammad Mikhail Bin Mazlan | Autonomous Sys-<br>tems Lead | _____     |
| Kannan s/o Rajamohan        | Data Manage-<br>ment Lead    | _____     |

Table 2: Debriefing Team Certification

**Report Version:** 1.0

**Debriefing Date:** February 7, 2026

**Methodology Status:** Validated and Recommended for Adoption

**Next Review:** 6 months after next project application

